

NED University of Engineering & Technology

Department of Computer and Information Systems Engineering

CS-221 Computer Organization and Design Lab

Session: 14 - Complex Engineering Activity (CEA) Date: _____

Student Name: _____ Roll No.: _____ Seat No.: _____ Section: _____

Lab 14 - Complex Engineering Activity (CEA)

Simulation of Single-Cycle RISC-V Datapath using Logisim

1. Objective

To simulate the complete single-cycle RISC-V datapath by integrating the modules developed in Lab Sessions 8-12: Register File, Instruction Fetch unit, ALU, Data Memory, Main Control Unit, and ALU Control Unit. The objective is to verify correct execution of R-type, I-type, S-type, and B-type RISC-V instructions.

2. Problem Statement

The single-cycle RISC-V datapath shown in Figure 14.1 is implemented in Logisim by integrating all previously designed datapath components. The given RISC-V program is loaded into instruction memory, executed step-by-step, and verified through register and data-memory contents.

Figure 14.1 - Single-Cycle RISC-V Datapath

[Attach screenshot of complete simulated datapath here]

3. Design & Integration Approach

Register File: Contains thirty-two 32-bit registers with two read ports and one write port.

Instruction Fetch Unit: Uses a 32-bit Program Counter and instruction memory to fetch instructions sequentially.

ALU: Performs add, sub, AND, and OR operations using ALU control signals.

Data Memory: Supports load and store operations controlled by MemRead and MemWrite.

Main Control Unit: Generates ALUSrc, MemToReg, RegWrite, MemRead, MemWrite, Branch, and ALUOp signals.

ALU Control Unit: Decodes ALUOp, funct3, and funct7 fields to generate the ALU operation.

Immediate Generator: Extracts and sign-extends immediates for I-type, S-type, and B-type instructions.

Branch Logic: Uses the Branch signal and ALU Zero flag to decide whether the branch is taken.

4. Task 1 - Reference Address & Data Memory Initialisation

Reference address xxx loaded in t0: 0x00000653

Table 14.1 - Partial Main Memory View

Main Memory Address	Actual Address Used	Contents
xxx	0x00000653	0x0000FF99

Main Memory Address	Actual Address Used	Contents
xxx + 4	0x00000657	0x0000DD88
xxx + 8	0x0000065B	0x0000CC77

Screenshot - relevant data memory portion:

[Attach screenshot of data memory initialisation here]

5. Task 2 - Instruction Translation & Instruction Memory

Table 14.2 - Code Snippet with Machine Code

Code Snippet	Machine Code Binary - 32 bits	Machine Code Hex
lw s1, 0(t0)	00000000000000101010010010000011	0x0002A483
lw s2, 4(t0)	0000000010000101010100100000011	0x0042A903
beq s1, s2, OR	00000001001001001000111001100011	0x01248E63
add t1, s1, s2	00000001001001001000001100110011	0x01248333
sw t1, 8(t0)	0000000011000101010010000100011	0x0062A423
sub t2, s1, s2	01000001001001001000001110110011	0x412483B3
sw t2, 12(t0)	0000000011100101010011000100011	0x0072A623
and t3, s1, s2	0000000100100100111111000110011	0x0124FE33
sw t3, 16(t0)	00000001110000101010100000100011	0x01C2A823
OR: or t4, s1, s2	00000001001001001110111010110011	0x0124EEB3
sw t4, 20(t0)	00000001110100101010101000100011	0x01D2AA23

Screenshot - instruction memory contents after loading program:

[Attach screenshot of instruction memory here]

6. Task 3 - Expected Register Contents After Execution

Table 14.3 - Register Contents

Register	Initial Contents	Expected Final Contents
t0	0x00000653	0x00000653
s1	0x00000000	0x0000FF99
s2	0x00000000	0x0000DD88
t1	0x00000000	0x0001DD21
t2	0x00000000	0x00002211
t3	0x00000000	0x0000DD88
t4	0x00000000	0x0000FF99

Analysis: s1 = 0x0000FF99 and s2 = 0x0000DD88, so s1 != s2. Therefore, beq is not taken and the program executes sequentially.

Computed values: t1 = 0x0001DD21, t2 = 0x00002211, t3 = 0x0000DD88, t4 = 0x0000FF99.

7. Task 4 - Single-Step Simulation & Results

The program was executed instruction-by-instruction using Logisim clock stepping. Probes were used to observe PC, ALU output, memory data, register values, and control signals.

Step 1 - lw s1, 0(t0): ALU computes address t0 + 0. Data memory returns 0x0000FF99, which is written to s1.

Step 2 - lw s2, 4(t0): ALU computes address t0 + 4. Data memory returns 0x0000DD88, which is written to s2.

Step 3 - beq s1, s2, OR: ALU subtracts s1 - s2. Since the result is not zero, Zero = 0 and the branch is not taken.

Step 4 - add t1, s1, s2: ALU adds s1 and s2. The result 0x0001DD21 is written to t1.

Step 5 - sw t1, 8(t0): ALU computes t0 + 8 and stores t1 at this data-memory address.

Step 6 - sub t2, s1, s2: ALU subtracts s2 from s1. The result 0x00002211 is written to t2.

Step 7 - sw t2, 12(t0): ALU computes t0 + 12 and stores t2 at this data-memory address.

Step 8 - and t3, s1, s2: ALU performs bitwise AND. The result 0x0000DD88 is written to t3.

Step 9 - sw t3, 16(t0): ALU computes t0 + 16 and stores t3 at this data-memory address.

Step 10 - or t4, s1, s2: ALU performs bitwise OR. The result 0x0000FF99 is written to t4.

Step 11 - sw t4, 20(t0): ALU computes t0 + 20 and stores t4 at this data-memory address.

Final Data Memory Contents

Address	Final Contents
t0 + 0	0x0000FF99
t0 + 4	0x0000DD88
t0 + 8	0x0001DD21
t0 + 12	0x00002211
t0 + 16	0x0000DD88
t0 + 20	0x0000FF99

Screenshot - register file contents after full program execution:

[Attach screenshot of final register contents here]

Screenshots - various stages of program execution:

[Attach screenshots at various stages of execution here]

8. Observations & Discussion

The datapath correctly executed load, store, branch, and R-type instructions. Since s1 and s2 were different, beq was not taken. The ALU correctly performed addition, subtraction, AND, and OR operations. Store instructions wrote computed values to data memory. The Main Control Unit and ALU Control Unit generated correct control signals and selected the proper ALU operations.

The Immediate Generator correctly produced immediate values for lw, sw, and beq instructions, confirming correct integrated datapath behavior.

9. Conclusion

The single-cycle RISC-V datapath was successfully simulated in Logisim by integrating the Register File, Instruction Fetch Unit, ALU, Data Memory, Immediate Generator, Main Control Unit, and ALU Control Unit. The final register and memory values matched the expected results, confirming the proper working of the complete single-cycle RISC-V processor datapath.

CEP Rubrics

For instructor use

Criterion	Score
CPA-1: Depth of Analysis	/4
CPA-2: Fundamental Knowledge	/4
CPA-3: Consequence of Design Change	/4
Uniqueness of Design	/4
Quality of Implementation	/4
TOTAL	/20

Remarks: _____

Instructor's Signature & Date: _____